TONOTOPY IN AN OSCILLATOR BANK AND THE INFLUENCE OF PARAMETER ADAPTATION

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Overview

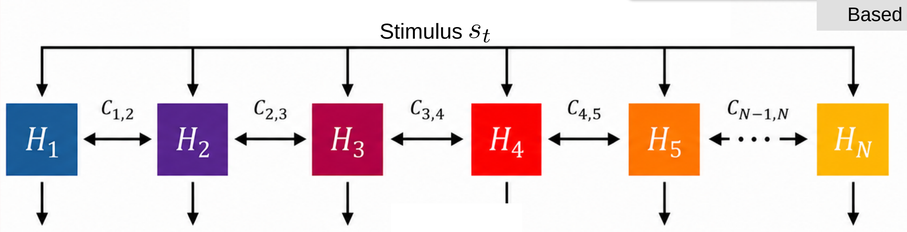
Oscillator-Bank Model: Introduction of a reduced oscillator representation derived from a transmission-line model
Single-Tone Tonotopy: Analysis of oscillation-frequency responses across the bank for different pure-tone stimuli

Two-Tone Tonotopy: Analysis of frequency responses to two pure tones and their effects on tonotopy, including the influence of local parameter adaptation.

Oscillator-bank model

Transmission-Line Model

Based on Zweig (1991)

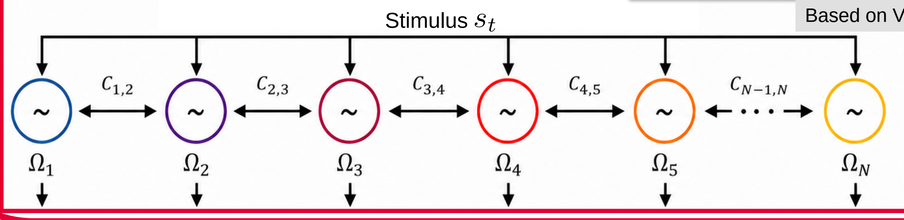


$$v_t = \dot{x}_t$$

$$\dot{v}_t = -\frac{k}{m}x_t - \frac{d}{m}v_t - \frac{1}{m}(\Psi(0) + \dot{\Psi}(0)x_{t-\tau} + \frac{\ddot{\Psi}(0)}{2}x_{t-\tau}^2)x_{t-\tau} + s_t$$

Oscillator Bank

Based on Vilfan&Duke (2008)



$$\eta_t = x_t - j\frac{1}{\omega}v_t$$

$$\dot{\eta}_t = (\mu + j\omega_0)\eta_t - (\alpha + j\beta)\left|\eta_t\right|^2\eta_t + s_t$$

x_t	:	Displacement
v_t	:	Velocity
η_t	:	Complex oscillator state
Ψ	:	Active-feedback function
m	:	Effective mass
k	:	Stiffness
d	:	Damping coefficient
ω_0	:	Natural angular frequency
μ	:	Linear dynamics parameter
α	:	Nonlinear dynamics parameter
β	:	Asynchronicity parameter
$C_{i,j}$	:	Coupling function between oscillator i and j
Ω_n	:	Oscillator index n

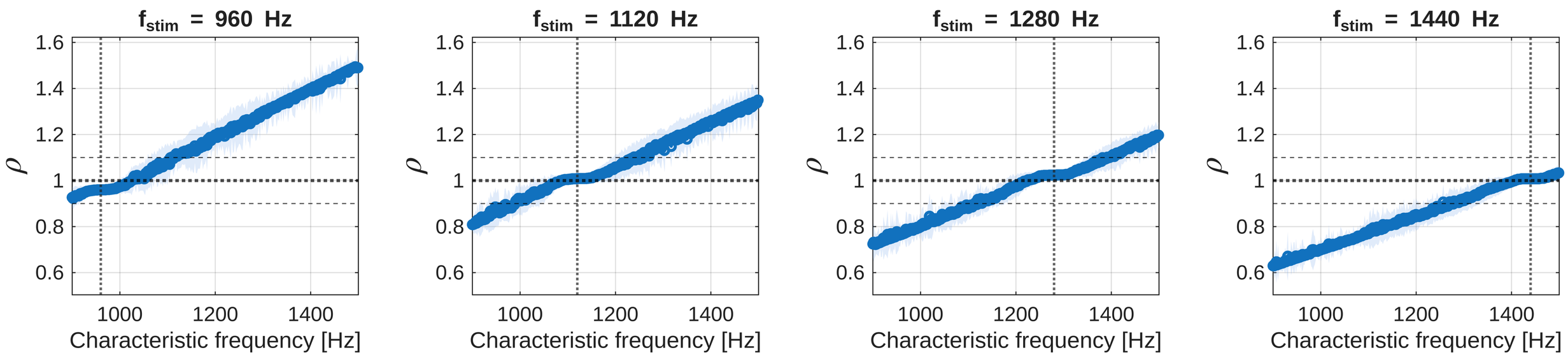
Tonotopy (Single Tone)

- **Stimulus**: Single pure tone, different frequencies, constant amplitude
- **Oscillator-bank**: 854 oscillators frequencies between 400Hz and 2kHz spaced based on Greenwoods map
- **Uniform parameters**: μ=0, α=1, β=1

$$\rho_n = \lim_{T\to\infty}\frac{\phi_n(t_0+T) - \phi_n(t_0)}{2\pi f_{\text{stim}}T}$$

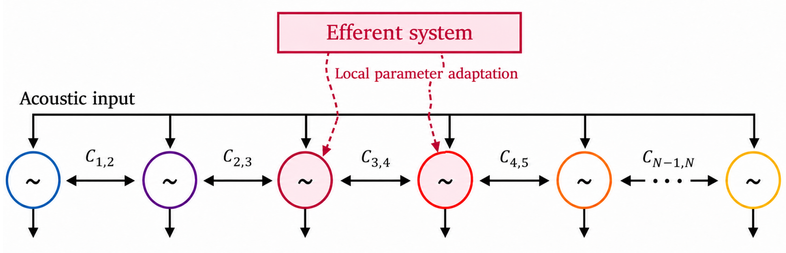
$$\rho_n = 1 \iff \overline{f}_n = f_{\text{stim}}$$

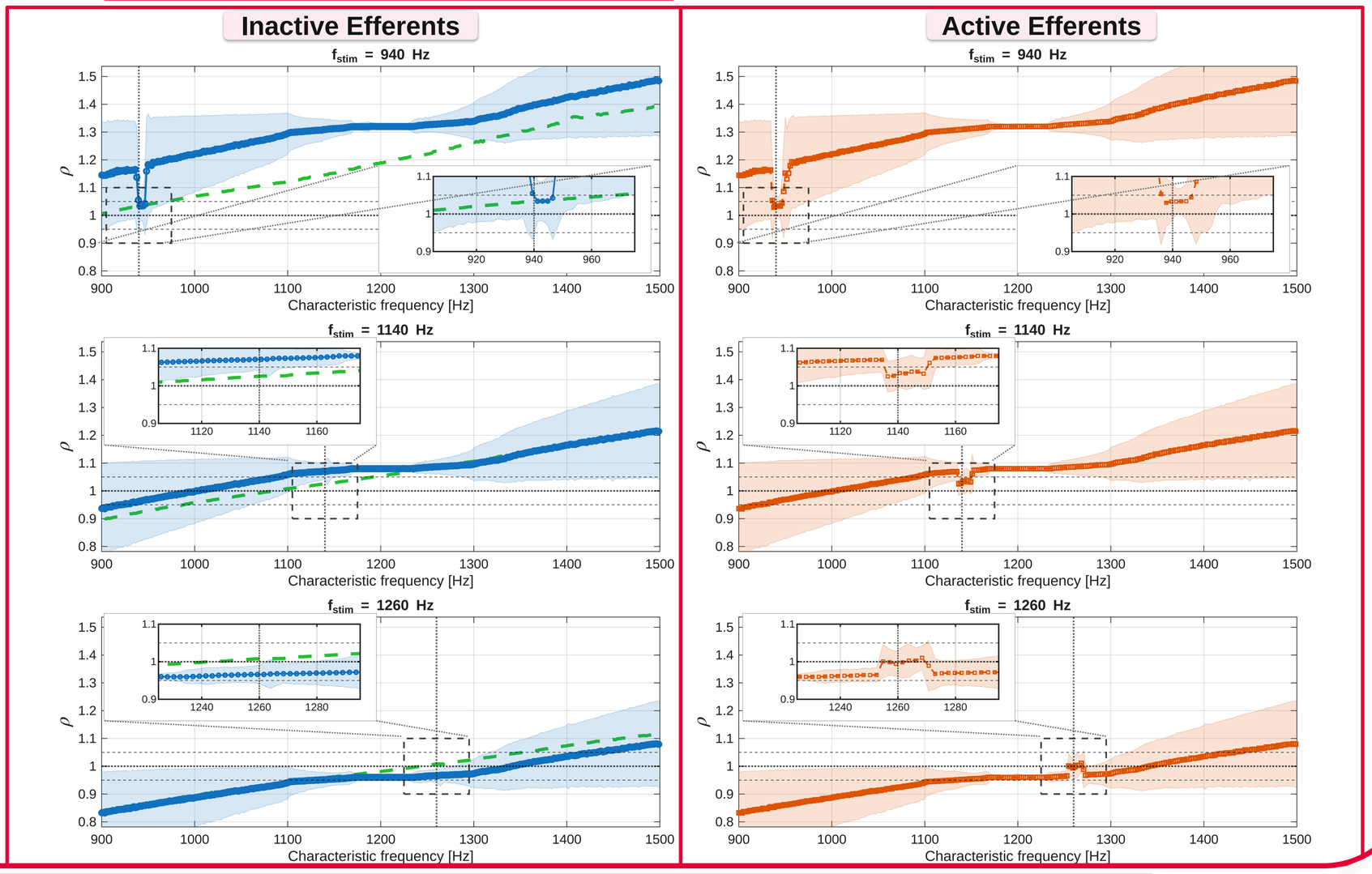
ρ_n	:	Rotation number of oscillator n
$\phi_n(t)$	:	Unwrapped phase of oscillator n
f_{stim}	:	Stimulus frequency
$\overline{f}_n$	:	Mean oscillation frequency
t_0	:	Start of analysis after transients
T	:	Observation duration



Masker Effects on Tonotopy

- **Target**: Pure tone, different frequencies, constant amplitude
- **Masker**: Simultaneous pure tone at 1.2kHz and, 84 dB above the stimulus level
- **Oscillator-bank**: 854 oscillators frequencies between 400Hz and 2kHz spaced based on Greenwoods map
- **Green:** Response to target without masker
- **Blue**: Inactive efferents α= 1, β=1
- **Red**: Active efferents with Local parameters α=0.1, β=0.1 near the stimulus frequency (10 oscillators); remaining oscillators unchanged





Conclusion

- **Stimulus-dependent tonotopy:** The stimulus shapes the oscillation frequencies across the bank, rather than activating only the oscillator with the closest characteristic frequency. Competing tones can therefore produce masking.
- **Adaptable frequency selectivity:** Local parameter changes can improve separation of a target from a masker, suggesting a possible role for efferent adaptation.